Public Economics (ECON 131)
Section #8: Externalities

GSI: Steve Ramos

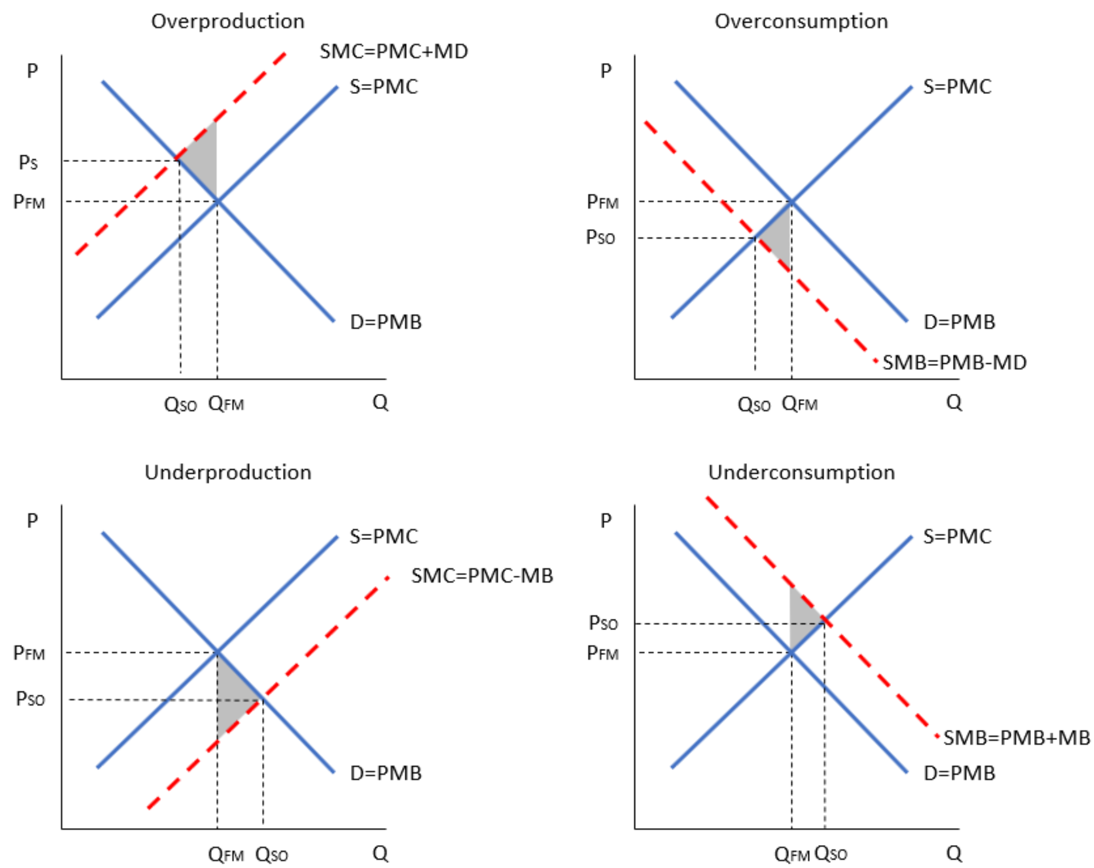
Contents

1 Externalities

1.1 Key concepts

- **Market failure:** A problem that causes the market economy to deliver an outcome that does not maximize efficiency.
  - Externalities, asymmetric information, individual failures
- **Externality:** An externality arises when the action of an agent directly makes other agents better off (positive externality) or worse off (negative externality), with no costs or benefits to the acting agent.
- Externalities can occur because of production or consumption
- Private vs. social marginal cost:
  - Private marginal cost (PMC): is the direct marginal cost of producing/consuming an additional unit of a good
  - Social marginal cost (SMC): is the PMC + marginal damages (MD), ie the PMC + any additional costs associated with production/consumption that are imposed on others in the economy

- In a free market, equilibrium where PMB = PMC
- Social optimum at SMB = SMC
- If PMB ≠ SMB and/or PMC ≠ SMC ⇒ private market leads to inefficient outcome



- We say an agent **internalizes the externality** when, due either to private negotiations or government actions, the price of an action fully reflects the external costs or benefits of that action.
  - Private negotiations can bring result in socially optimal quantity as long as (a) there are well-defined property rights (b) costless bargaining, and (c) competitive markets (**Coase theorem**—but see lecture slides for some limitations)
  - Public policy can enter when Coase conditions are not met:
    - price policy (tax or subsidy)
    - quantity regulation (command and control or tradable permits)

1.2 Practice Problems

1.2.1 Based on Gruber Ch.05, Q.13

Suppose that demand for a product is $Q = 1,000 - 4P$ and supply is $Q = -200 + 2P$. Furthermore, suppose that the marginal external damage of this product is \$6 per unit.

(a) How many more units of this product will the free market produce than is socially optimal?

(b) Calculate the deadweight loss associated with the externality.

1.3 Additional Practice Problems

1.3.1 Based on Gruber Ch.05, Q.14

The marginal damage averted from pollution cleanup is $MD = 200 - 5Q$. The marginal cost associated with pollution cleanup is $MC = 20 + Q$.

(a) What is the optimal level of pollution reduction?

(b) Show that this level of pollution reduction could be accomplished through taxation. What tax per unit would generate the optimal amount of pollution reduction?

1.3.2 Gruber Ch.05, Q.4

In the midwestern United States, where winds tend to blow from west to east, states tend to more easily approve new polluting industries near their eastern borders than in other parts of the state. Why do you think this is true?